

Code-Layout, Readability and Reusability

Date	28 June 2024
Team ID	SWTID-2026-1236
Project Name	Credit Card Approval Eligibility
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Proper indentation is maintained throughout the code.
2	Proper File Structure	Files and folders are logically organized	Yes	Frontend, backend, machine learning model, and database files are organized into separate folders.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables are named clearly to reflect their purpose.
4	Function / Method Names	Functions are descriptively named	Yes	Functions have descriptive names that indicate their functionality.
5	Code Comments	Inline and block comments explain logic	Partial	Comments are added to explain important sections of the code.
6	Modular Design	Code is split into reusable functions/modules	Yes	The application is divided into reusable modules such as frontend, backend, AI model, and database.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Duplicate and unnecessary code is removed to improve maintainability.
8	Error Handling	Exceptions and errors are handled carefully	Yes	Input validation and exception handling are implemented to manage errors gracefully.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	User Interface	HTML, CSS, JavaScript	Used across all web pages	High
2	Flask Backend	Python (Flask)	Handles form submission and communication with the AI model	High
3	AI & ML Prediction Model	Scikit-learn	Used to predict customer eligibility	High
4	Input Validation Module	JavaScript, Python	Validates user input before prediction	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project structure with separate frontend, backend, ML, and database modules.
Readability	5	Clear naming conventions, indentation, and comments improve readability.
Reusability	5	Modular components can be reused in future AI & ML web applications.
Documentation / Comments	4	Code is documented with comments where necessary.
Overall Score	4.75/5	The project follows good coding practices and is easy to maintain and extend.